

The nonlinear tuning technique of a DFB laser using frequency predistortion procedures

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ABSTRACT

The frequency modulation continuous wave (FMCW) light detection and ranging (LIDAR) has drawn enormous attention and intense effort. In this paper a nonlinear tuning technique for a DFB laser which is driven by a home-made LD driver is proposed and developed experimentally for generation of linearly chirped light which may be applied in FMCW LIDAR systems. A frequency predistortion procedure based on nonlinear tuning technique can be applied to the LD driver which could be controlled by a field-programmable gate array (FPGA) module. And the frequency predistortion procedure can be used iteratively until the desired linearity of the chirped optical waves is achieved before the PLL is introduced in. This technique can be seen as an effective way to increase the acquisition bandwidth of the PLL significantly so that the trade-off between large acquisition frequency and narrow bandwidth of the PLL is no longer a problem.

Keywords: FMCW, frequency-swept DFB laser, predistortion processing, frequency-swept nonlinearity

1. INTRODUCTION

FMCW used in LIDAR is an electromagnetic wave which carrier signal frequency changes linearly with time^[1-2]. LIDAR technology has the advantages of high resolution, high precision, non-contact and strong anti-jamming ability etc. LIDAR has great potential in the fields of driverless vehicles^[3]. The frequency variation form of FMCW usually includes sawtooth wave and triangle wave. In the distance measurement technology, FMCW LIDAR transmits linear frequency modulation continuous wave signal to the target and receives the echo signal reflected by the target. Through coherent detection, it measures the frequency difference introduced by the distance delay between the echo signal and the transmitted signal to solve the distance of the target. However, in order to obtain high measurement accuracy and ranging resolution^[4], it is necessary to generate a linear frequency modulated continuous optical signal with high linearity.

Based on the control of injection current (or change of operating temperature), the frequency modulation light source of distributed feedback semiconductor laser (DFB) has been widely used for its advantages of simple control^[5], low price and small overall system. However, the inherent frequency-sweep nonlinearity of DFB laser restricts the application and development of LIDAR technology, especially for ranging measurement. Therefore, it is necessary to correct the frequency modulation nonlinearity of DFB lasers. The system of an emitting light source based on a DFB laser (shown in Figure 1) is a practical scheme to be completed for LIDAR application. The system, which can achieve high-precision nonlinear correction of DFB laser, combines two parts that are a current predistortion unit and a photoelectric feedback loop, both for nonlinear correction, actually^[6].

As illustrating in Fig.1, the main mechanism of the closed-loop correction is to use the time-delay self-heterodyne method to build a photoelectric feedback loop to correct the output signal frequency of the DFB laser^[7]. By building a Mach Zehnder interferometer (MZI)^[8], the absolute values of the delay time of the optical signal and the frequency modulation slope of the DFB laser are converted into the frequency of the beat signal of the photodetector (PD) which is shown in Fig.1. Comparing the frequency of the PD beat signal with that of a standard reference signal through a mixer, a negative feedback system is established to compensate the frequency difference between the beat signal and standard reference signal. The frequency of laser output signal changes linearly with time by stabilizing the frequency of MZI

beat signal to the frequency of the microwave reference source. Compared with other correction methods, the closed-loop correction method can achieve high-precision nonlinear correction and dynamic real-time correction.

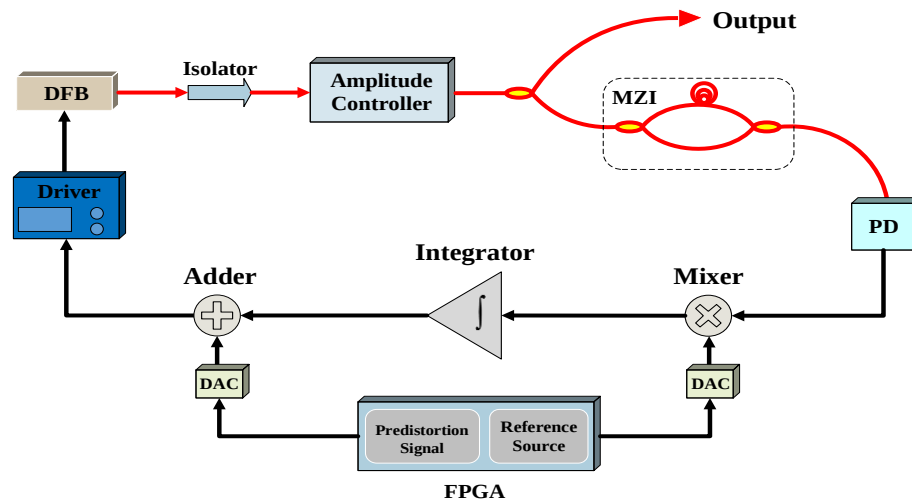


Figure 1. A Linear frequency-swept photoelectric phase-locked closed-loop system with a pre-distortion current injection open-loop unit.

To preprocess the frequency modulation nonlinearity of the DFB laser before activating the closed loop correction, the function of open loop correction unit is used, and it can also be seen as an independent non-linear correction subsystem. There is a reason for initiating the open loop correction unit at the first step. Due to the limited capture range of the photoelectric phase-locked loop ^[9], once the frequency of the error signal in the phase-locked loop exceeded the synchronizing band of the phase-locked loop, the lock-out phenomenon might occur. Therefore, although the non-linear correction accuracy of the open-loop correction is low, it still could effectively reduce the non-linearity of the sweep frequency of the laser, and at the same time, it may expand the capture bandwidth of the phase-locked loop in the closed-loop correction at the second step.

For the open loop correction method, a specific modulation current signal (i.e. predistortion current) is input into the DFB laser to make the frequency modulation characteristic of the laser to be tuned approximately linearly with time. The methods for obtaining the predistortion current include table-lookup method and iteration method ^[10]. The iterative algorithm is most commonly used to generate the predistortion current. This method usually uses a transfer function to inversely resolve the predistortion waveform function, and then a waveform generator generates a specific waveform current signal according to the predistortion waveform function, and injects the current signal into the DFB laser to reduce the frequency modulation nonlinearity of the DFB laser.

The open-loop correction is simple in system structure and easy to realize, and the frequency modulation nonlinearity of the laser can be greatly reduced through correction algorithms. If a reasonable or moderate accuracy for LIDAR ranging is required, the frequency-modulated signal generated only by open-loop correction alone may meet the requirement of the LIDAR application scenario. And as an independent non-linear correction method, the open-loop iterative predistortion is also chosen in most of non-linear correction schemes. Furthermore, for achieving high-precision non-linear correction, open-loop correction is also a necessary prerequisite for realization of a closed-loop correction in a FMCW LIDAR light source. In this paper, the frequency modulation nonlinear correction technology of a DFB laser based on current predistortion is studied. A simulation model for pre-correction of a DFB laser is established, and the procedure of laser frequency modulation nonlinear correction based on the predistortion iterative algorithm is demonstrated.

2. PRINCIPLE OF ITERATIVE PREDISTORTION

2.1 Introduction of experimental system

The current modulation frequency response of DFB laser is related to carrier concentration and temperature. When the injection current changes, the carrier concentration in the active region also changes, and then the refractive index of the active region is changed, so that the laser frequency is modulated, but this process is accompanied by the change of temperature of the active region. The larger the injection current, the more obvious the thermal effect of the carrier is. For current modulated semiconductor lasers, there is no situation that only a single modulation mechanism exists, so frequency modulation nonlinearity is a complexed performance for a DFB laser. Figure 2 shows the block diagram of the experimental system of laser frequency modulation nonlinear correction based on injection current predistortion technology.

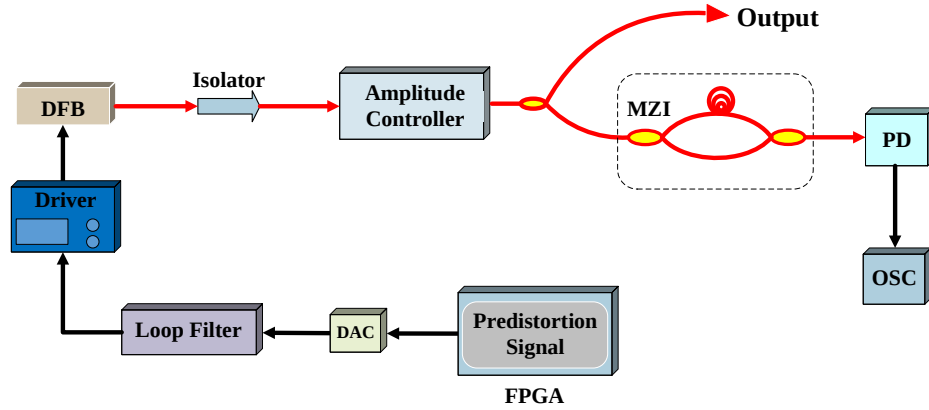


Figure 2. Block diagram of LFM optical system based on injection current predistortion.

The predistortion signal generated by FPGA drives the laser to sweep frequency linearly. The 10% output power of the DFB laser is injected into an unbalanced MZI through a 90/10 fiber coupler. The optical signal of one arm is delayed by 20 ns relative to the other, and the beat signal of the two arms is obtained and mixed at a photodetector (PD). The electric beat signal can be collected and recorded by an oscilloscope (OSC) for post processing which is described in the next session. A FPGA chip can complete the calculation of the beat signal and output the predistortion current to realize the first order correction. When the 1st-order predistortion current is injected into the laser again, and a 2nd-order predistortion current can be obtained by calculating with a new beat signal recorded by the OSC. This is the iterative method to correct the frequency-swept nonlinearity of the DFB laser. The number of iterations of this method is limited, and it is not allowed to iterative all the time. It would stop once the frequency modulation linearity of the laser reaches the requirement of applications.

2.2 Theoretical analysis

In a sweeping period, the instantaneous frequency of the sweep laser can be expressed as:

$$\omega(t) = \omega_0 + K_{SCL} [i(t)] \cdot i(t) \quad (1)$$

where $K_{SCL} [i(t)]$ is the current dependent gain or current frequency modulation coefficient in the nonlinear frequency modulation response. In order to characterize the chirp rate, the spectrum of the photocurrent of the PD beat signal can be obtained. The relationship between the frequency of the photocurrent and the derivative of the instantaneous frequency of the laser is shown in equation (2):

$$\omega_{PD} = \tau \frac{d\omega(t)}{dt} \quad (2)$$

where τ is the delay of MZI. From equation (1), we can get equation (3) below:

$$\begin{aligned}
\omega_{PD}(t) &= \tau \frac{d\omega(t)}{dt} = \frac{di(t)}{dt} \left(\tau K_{SCL} + \tau i \frac{dK_{SCL}}{di} \right) \\
&= \frac{di(t)}{dt} \left(K_{SCL} + i \frac{dK_{SCL}}{di} \right) \cdot \tau \\
&= \frac{di(t)}{dt} \cdot F_{dist}(i)
\end{aligned} \tag{3}$$

where $\frac{di(t)}{dt} \left(K_{SCL} + i \frac{dK_{SCL}}{di} \right)$ is the frequency sweep rate. The nonlinear function can be obtained from equation (3) and shown in equation (4) below:

$$F_{dist}(i) = \frac{\omega_{PD}(t)}{\frac{di(t)}{dt}} \tag{4}$$

where $F_{dist}(i)$ is the nonlinear function.

We can get the differential equation of modulation current from equation (3):

$$\frac{di(t)}{dt} = \frac{\tau}{F_{dist}(i)} \frac{d\omega(t)_{desired}}{dt} \tag{5}$$

where $\omega(t)_{desired}$ is the instantaneous frequency of the desired linear sweep light, we can reduce the sweep nonlinearity of the laser by solving the current obtained from equation (5). However, the model is not perfect. In order to further reduce the laser frequency-sweep nonlinearity, we use the iterative predistortion method of the model to make the laser frequency-sweep characteristics close to the expected performance.

The iterative predistortion means that the new function is recalculated by the predistortion current obtained previously, and then the next approximation is calculated by solving equation (5) again. This process is repeated until the laser output frequency-sweep light is close to the ideal linearity. We can get the differential equation of the n^{th} -order predistortion current by equation (3) and equation (5), as shown in equation (6):

$$\frac{di(t)_n}{dt} = \frac{\tau}{F_{dist}(i)_{n-1}} \frac{d\omega(t)_{desired}}{dt} \tag{6}$$

$$F_{dist}(i)_{n-1} = \frac{\omega_{PD}(t)_{n-1}}{\frac{di(t)_{n-1}}{dt}} \tag{7}$$

where $F_{dist}(i)_{n-1}$ is the $(n-1)^{\text{th}}$ -order nonlinear function. After doing the predistortion processing several times, the difference between the two adjacent predistortion currents is very small. At this time, the iteration can be completed. Because the ability of the predistortion iteration algorithm to improve the linearity has reached its limit.

3. SIMULATION RESULTS OF ITERATIVE PREDISTORTION

In this paper, according to figure 2, simulation software is used to simulate the system, and a specific relationship of output frequency of the DFB laser is described by the function with time below:

$$\omega(t) = \omega_0 + \sqrt{t} \tag{8}$$

In order to explain the model conveniently, a period of 20 μs is taken as an example. The frequency of the model changes 10 GHz in a period, and the first injected current increases linearly from 70 mA to 80 mA in a period. Using the

predistortion iteration theory in Section 2.2, the relationship between the injection current and the laser output frequency is established. According to the simulation operation of the iterative algorithm, after repeated experiments, we found that the correction effect of four times iterations is the best, and the continuous iteration improvement is very little. The effect of sweep nonlinear correction is not obvious, which shows that the iteration is convergent. The results of four times iterations are considered as the optimal results of open-loop correction. Figure 3 shows the result of iterative predistortion. Each predistortion will produce a chirp which is closer to the desired frequency modulation characteristic, so that a chirp with enough linearity will be obtained at the fourth order.

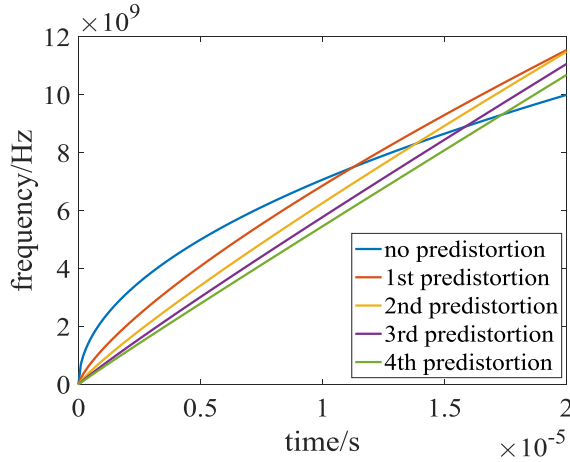


Figure 3. Iterative predistortion spectrogram.

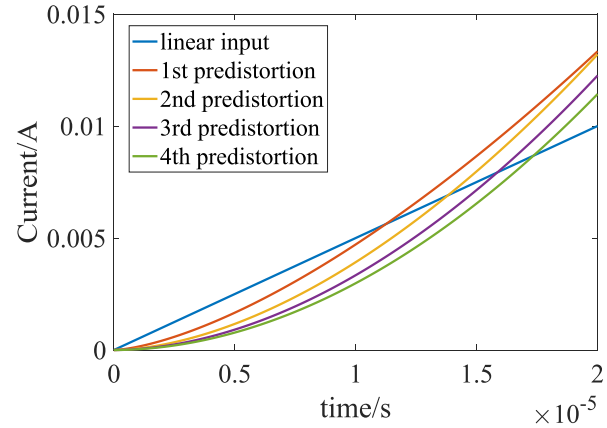


Figure 4. Calculated driving current to the laser for different orders of predistortion.

The linearity R^2 of the output frequency curve with time of the DFB laser is 0.957123, 0.991440, 0.998057, 0.999536, 0.999887, respectively. It can be seen that after iterative predistortion, the linearity of frequency modulation of the laser has been significantly improved.

Figure 4 shows the driving current of the laser before and after predistortion. It can be seen from the figure that the modulation range of the current after predistortion will be expanded to a certain extent, but the modulation range of the current will be close to the range of the current without predistortion as the number of iterations increases.

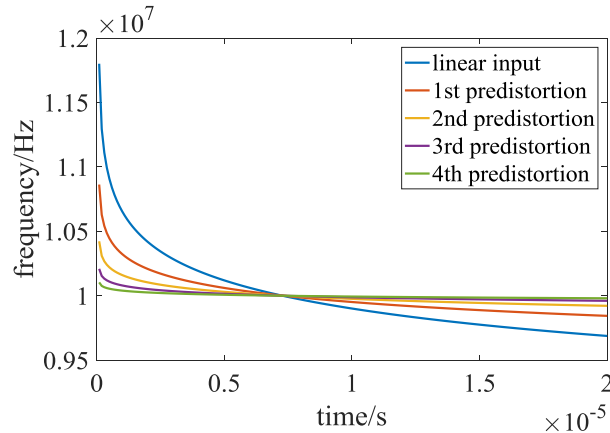


Figure 5. The spectrum of the PD beat signal for linear and predistortion.

Figure 5 shows the spectrum of PD beat signal before and after predistortion. It can be seen from the figure that after four iterations, the spectrum width of PD beat signal is greatly narrowed, and the spectrum of beat signal is close to the theoretical value, indicating that the frequency modulation nonlinearity of laser is reduced.

4. CONCLUSIONS

The paper analyzes the generation mechanism of frequency-sweep nonlinearity of DFB laser controlled by injection current, proposes a frequency-sweep nonlinearity correction method based on the drive current predistortion, and builds up a linear sweep frequency system based on the current predistortion correction principle to reduce the frequency error range caused by sweep nonlinearity. By establishing the time-frequency characteristic model of the laser and using software to simulate the system according to the principle of iterative predistortion, the simulation results show that the sweep nonlinearity of the laser can be corrected by the method in this paper, and the linearity of the DFB laser after four times iterations reaches 0.99887, which achieves the expected levels. It can keep the PLL in a reasonable large capture bandwidth to make the whole system functioning, and lay a foundation for further improvement of frequency modulation nonlinear correction of DFB laser by using a PLL technique.

5. ACKNOWLEDGMENTS

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